

WTO Accession, PNTR, and U.S. Import Quantities and Prices

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Outline

- Background and research question
- Data and empirical strategy
- Current results and robustness
- Next steps

Background: U.S.–China Trade Relations

- **1980:** U.S. grants Communist China low tariff rates (MFN status) for the first time, but as a requires annual renewal reflecting Cold War era distrust.
- **1980s:** Renewals are routine. China's share of U.S. imports is small.
- **1989:** Tiananmen Square crackdown makes renewal politically contentious. Congress begins using the annual vote as leverage on human rights and Taiwan.
- **1990s:** Legislation to block China's NTR is introduced every year from 1989–1999. Yet U.S. imports from China keep growing: from \$51.5B (1996) to \$102B (2001).

Policy Timeline

- **Pre-2001:** China receives low tariff rates (NTR) each year, but Congress must vote to renew. If renewal fails, tariffs revert to Smoot-Hawley rates from 1930 (an average jump from 4% to 37%).
- **October 2000:** Congress passes the U.S.–China Relations Act, granting China **Permanent Normal Trade Relations (PNTR)**, conditional on WTO accession.
- **December 2001:** China officially joins the WTO.
- **January 2002:** Permanent low tariffs take effect. The annual renewal process is eliminated.

Critical insight

The actual tariff rates mostly **did not change**: China was already receiving the low NTR rate every year. What changed was the **permanent elimination of uncertainty** about whether those low rates would continue.

Research question

How did the resolution of U.S. trade-policy uncertainty around China starting in 2002 affect:

- the volume of imports from China into the U.S., and
- the prices U.S. buyers paid for Chinese imports?

Why did uncertainty matter differently across products?

- Every product had two legislated tariff rates: a low NTR rate and a high non-NTR rate.
- The **NTR gap** = non-NTR rate – NTR rate measures how much a product's tariff *would have jumped* if Congress didn't renew.
- Products with large gaps faced more risk → importers could be heavily impacted.
- Products with small gaps faced less risk → less impact by the annual vote.

Identification Strategy: Difference-in-Differences

Core idea

Compare import growth for **high-gap** products (more exposed to uncertainty) vs. **low-gap** products (less exposed), before and after WTO entry.

Treatment exposure (measured in 1999, before WTO negotiations):

$$\text{gap_pre}_p = (\text{non-NTR tariff})_p - (\text{NTR tariff})_p$$

Products above the median gap $\rightarrow$ “high-gap” (treatment); below $\rightarrow$ “low-gap” (control).

Data and Sample

Data sources

- USITC monthly imports by product and HTS.
- Tariff schedule information used to construct product-level `gap_pre`.

Panel and outcomes

- Product-month panel .
- Outcomes: `ln_customs_value`, `ln_first_unit_qty`, `ln_unit_value`.

Baseline Specification

Two-way FE DiD

$$Y_{pt} = \beta (\text{HighGap}_p \times \text{Post}_t) + \delta_p + \lambda_t + \varepsilon_{pt}$$

- Y_{pt} : log import value, log quantity, or log unit value.
- δ_p : product fixed effects.
- λ_t : time fixed effects.
- SEs clustered at product level.

Main DiD Results (Current Outputs)

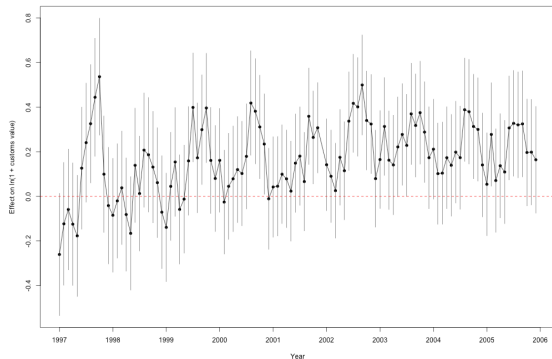
Binary treatment (HighGap x Post) with product and date FE

- $\ln_customs_value$: $\hat{\beta} \approx 0.121$ (s.e. 0.031), positive and significant.
- $\ln_first_qty$: $\hat{\beta} \approx 0.145$ (s.e. 0.036), positive and significant.
- $\ln_unit_value$: $\hat{\beta} \approx -0.024$ (s.e. 0.017), negative but not strongly significant in FE spec.

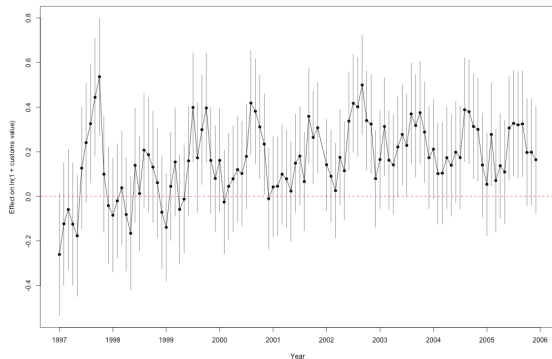
Interpretation

- High-gap products show stronger post-policy growth in value and quantity.
- Price effects are weaker in the binary split and clearer in continuous treatment.

Event Study: Value and Quantity

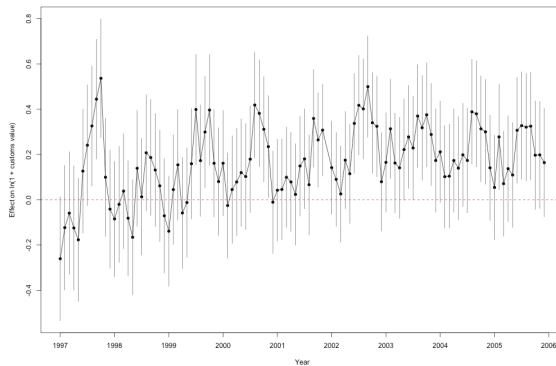


Value event study



Quantity event study

Event Study: Unit Value (Price Proxy)



Continuous-treatment pre-trend test for unit value: $p \approx 0.081$ (marginal).

Takeaways and Next Steps

Takeaways

- Evidence supports a post-policy increase in import value and quantity for high-gap products.

Next steps

- empty

Questions